

EE3621: Digital Signal Processing

Lecture 24: Design of IIR filters from analog filters - Impulse invariance (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** Foundations of IIR Filter Design: Comparison between FIR and IIR filters. Why digital IIR filters are synthesized by transforming continuous-time analog prototypes ($H_a(s) \rightarrow H(z)$).
 - **06:00 – 16:00 (10 mins):** The Principle of Impulse Invariance: Sampling the continuous-time impulse response $h_a(t)$ at $t = nT_s$. Derivation of the discrete impulse response $h[n] = T_s h_a(nT_s)$ and the fundamental pole mapping rule $s_k \rightarrow z_k = e^{s_k T_s}$.
 - **16:00 – 24:00 (8 mins):** Partial Fraction Expansion Synthesis: Expressing $H_a(s) = \sum_{k=1}^N \frac{A_k}{s-s_k}$ and transforming directly into causal digital poles $H(z) = \sum_{k=1}^N \frac{A_k}{1-e^{s_k T_s} z^{-1}}$. Proof of stability preservation ($\text{Re}(s_k) < 0 \implies |z_k| < 1$).
 - **24:00 – 30:00 (6 mins):** The Aliasing Dilemma: Analysis of frequency relationship $\omega = \Omega T_s$. The periodic spectral superposition $H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega-2\pi k}{T_s}\right)$. Why impulse invariance is strictly limited to band-limited Lowpass/Bandpass filters and fails for Highpass and Bandstop filters.
 - **30:00 – 38:00 (8 mins):** Step-by-Step Worked Numerical Examples: Complete manual transformation of a 1st-order and 2nd-order analog Butterworth lowpass prototype into causal digital IIR filters with real biquad coefficients.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Explain** why IIR digital filters are designed via transformation from continuous-time analog filter prototypes.
2. **Formulate** the impulse invariance criterion: $h[n] = T_s h_a(nT_s)$.
3. **Prove** that every simple pole in the s -plane at $s = s_k$ maps directly to a discrete pole in the z -plane at $z = e^{s_k T_s}$.

4. **Demonstrate** that the left-half of the s -plane ($\text{Re}(s) < 0$) maps strictly to the interior of the unit circle ($|z| < 1$), guaranteeing stability.
 5. **Apply** the partial fraction expansion method to synthesize digital transfer functions $H(z)$ without computing convolution integrals.
 6. **Analyze** the origin and severity of aliasing distortion in impulse invariance design.
 7. **Deduce** why impulse invariance is entirely inappropriate for Highpass and Bandstop filter design.
 8. **Synthesize** 1st-order and 2nd-order digital Butterworth filters from analog prototypes through complete numerical calculations.
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2 2. MOTIVATION: DESIGNING IIR FILTERS FROM ANALOG PROTOTYPES

2.1 2.1 FIR vs. IIR Filtering Paradigms

In Lectures 21–23, we designed Finite Impulse Response (FIR) filters. FIR filters possess exact linear phase and guaranteed stability. However, achieving sharp transition roll-offs requires very high filter orders ($N = 50$ to 500), incurring substantial memory and arithmetic latency.

Infinite Impulse Response (IIR) filters, governed by the rational transfer function:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (1)$$

exploit recursive feedback (poles). Consequently, an IIR filter can meet extremely stringent magnitude specifications with a fraction of the filter order ($N = 2$ to 8) compared to an FIR filter.

2.2 2.2 Why Start from Continuous-Time Analog Filters?

Continuous-time analog filter theory (Butterworth, Chebyshev, Elliptic) is a mature, mathematically closed engineering discipline developed over a century. Closed-form polynomials exist that yield optimal, mathematically proven approximations to ideal brick-wall responses.

Therefore, the standard strategy for designing a digital IIR filter is:

1. Translate digital frequency specifications ($\omega_p, \omega_s, \delta_p, \delta_s$) into equivalent analog specifications (Ω_p, Ω_s).
2. Design an analog prototype filter $H_a(s)$ meeting those specifications.
3. Transform the continuous-time system function $H_a(s)$ into a causal discrete-time system function $H(z)$ via an algebraic or time-domain mapping.

Core Requirements for the Mapping Technique:

- **Preservation of Stability:** Poles in the stable left-half s -plane ($\text{Re}(s) < 0$) **MUST** map inside the unit circle of the z -plane ($|z| < 1$).
 - **Preservation of Frequency Characteristics:** The imaginary axis of the s -plane ($s = j\Omega$) must map onto the unit circle ($z = e^{j\omega}$) so that analog passbands and stopbands translate to digital passbands and stopbands.
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3. THE IMPULSE INVARIANCE METHOD: MATHEMATICAL FOUNDATIONS

3.1 The Time-Domain Invariance Criterion

The philosophy of the **Impulse Invariance Method** is to produce a digital filter whose discrete impulse response $h[n]$ is an exact sampled replica of the continuous-time analog impulse response $h_a(t)$:

$$h[n] = T_s h_a(nT_s), \quad n = 0, 1, 2, \dots \quad (2)$$

where T_s is the sampling interval (seconds). The scaling factor T_s is introduced to ensure that the digital filter's DC gain matches the analog filter's DC gain:

$$H(e^{j0}) = \sum_{n=0}^{\infty} h[n] = T_s \sum_{n=0}^{\infty} h_a(nT_s) \approx \int_0^{\infty} h_a(t) dt = H_a(0) \quad (3)$$

3.2 Mapping of Poles from s-Plane to z-Plane

Let the analog filter transfer function $H_a(s)$ be described by distinct poles s_k :

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \quad (4)$$

Taking the inverse Laplace transform of each simple pole term yields the causal analog impulse response:

$$h_a(t) = \sum_{k=1}^N A_k e^{s_k t} u(t) \quad (5)$$

Sampling this response at $t = nT_s$ and multiplying by T_s as per (2):

$$h[n] = T_s \sum_{k=1}^N A_k e^{s_k n T_s} u[n] = \sum_{k=1}^N (T_s A_k) (e^{s_k T_s})^n u[n] \quad (6)$$

Now, taking the unilateral z -transform of $h[n]$:

$$\begin{aligned} H(z) &= \sum_{n=0}^{\infty} h[n] z^{-n} = \sum_{n=0}^{\infty} \left[\sum_{k=1}^N T_s A_k (e^{s_k T_s})^n \right] z^{-n} \\ &= \sum_{k=1}^N T_s A_k \sum_{n=0}^{\infty} (e^{s_k T_s} z^{-1})^n \end{aligned} \quad (7)$$

Applying the standard geometric series summation $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$ (convergent for $|e^{s_k T_s} z^{-1}| < 1$):

$$\mathbf{H}(z) = \sum_{k=1}^N \frac{\mathbf{T}_s \mathbf{A}_k}{1 - e^{s_k T_s} z^{-1}} \quad (8)$$

Fundamental Operational Rule of Impulse Invariance:

$$\frac{\mathbf{A}_k}{s - s_k} \xrightarrow{\text{Impulse Invariance}} \frac{\mathbf{T}_s \mathbf{A}_k}{1 - e^{s_k T_s} z^{-1}} \quad (9)$$

Every analog pole at $s = s_k$ maps directly to a discrete digital pole at:

$$z_k = e^{s_k T_s} \quad (10)$$

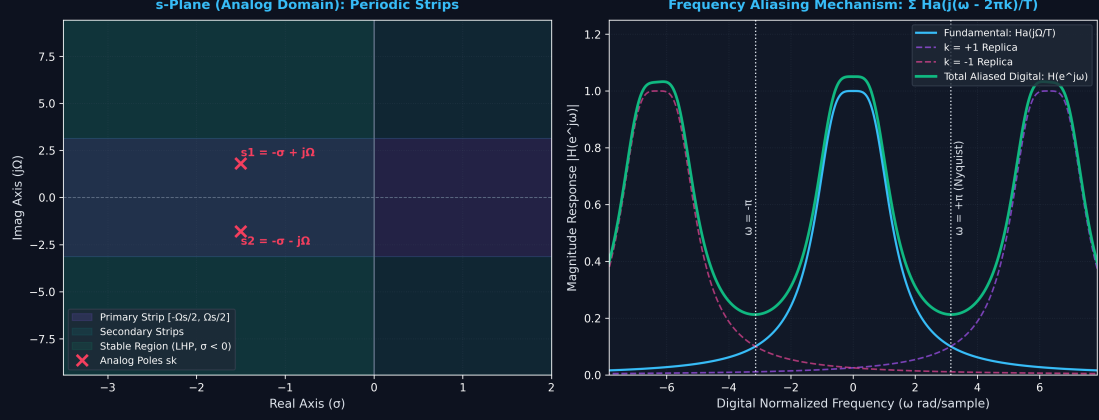


Figure 1: Impulse Invariance transformation. Left: Periodic horizontal strips of width $\Omega_s = 2\pi/T_s$ in the s -plane mapping into the unit circle in the z -plane. Right: Frequency aliasing caused by the periodic superposition of shifted analog spectra $\sum H_a(j(\omega - 2\pi k)/T_s)$.

3.3 Stability Mapping Analysis

Let an analog pole be written in rectangular form: $s_k = \sigma_k + j\Omega_k$. The corresponding digital pole is:

$$z_k = e^{(\sigma_k + j\Omega_k)T_s} = e^{\sigma_k T_s} \cdot e^{j\Omega_k T_s} \implies |z_k| = e^{\sigma_k T_s} \quad (11)$$

1. **Stable Analog Poles ($\sigma_k < 0$):** If the analog pole lies in the left-half of the s -plane, then $\sigma_k < 0 \implies |z_k| = e^{\sigma_k T_s} < 1$. The pole maps **strictly inside the unit circle!**
2. **Marginally Stable Poles ($\sigma_k = 0$):** If the analog pole lies on the $j\Omega$ -axis, $|z_k| = e^0 = 1$. The pole maps **on the unit circle**.
3. **Unstable Poles ($\sigma_k > 0$):** If the analog pole lies in the right-half plane, $|z_k| = e^{\sigma_k T_s} > 1$. The pole maps **outside the unit circle**.

Theorem: The Impulse Invariance method **unconditionally preserves system stability**. A stable continuous-time filter always yields a strictly stable discrete-time digital filter.

4. THE FREQUENCY ALIASING PHENOMENON

4.1 Frequency-Domain Mathematical Relationship

Because $h[n]$ is obtained by sampling $h_a(t)$ at uniform intervals T_s , the continuous-time Fourier transform $H_a(j\Omega)$ and the discrete-time Fourier transform $H(e^{j\omega})$ are related by the standard Poisson Summation formula:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega - 2\pi k}{T_s}\right) \quad (12)$$

where normalized digital frequency ω is related to analog angular frequency Ω by:

$$\omega = \Omega T_s \quad (13)$$

4.2 4.2 Origin of Severe Aliasing Distortion

Equation (12) shows that $H(e^{j\omega})$ is an infinite periodic superposition of replicas of $H_a(j\Omega)$ shifted by integer multiples of the sampling frequency $\Omega_s = \frac{2\pi}{T_s}$.

Unlike discrete signals which can be passed through continuous anti-aliasing lowpass filters before digitization, **an analog filter transfer function $H_a(s)$ is a continuous mathematical rational function whose frequency response extends to $\Omega \rightarrow \pm\infty$.**

- Because $H_a(j\Omega)$ cannot be strictly band-limited to $|\Omega| \leq \frac{\pi}{T_s}$, the high-frequency tails fold back into the baseband $[-\pi, \pi]$!
- This spectral foldover causes ****irreversible aliasing distortion**** in the passband and transition band.

4.3 4.3 Why Impulse Invariance Fails for Highpass and Bandstop Filters

For an analog Highpass filter, $|H_a(j\Omega)| \rightarrow 1$ as $\Omega \rightarrow \infty$. When transformed via impulse invariance:

- The non-zero high-frequency response aliases across all periodic copies $k = \pm 1, \pm 2, \dots$
- These infinite overlapping tails superimpose at DC ($\omega = 0$), causing the digital filter's response to be non-zero at DC!
- **Fatal Consequence:** The designed filter completely fails to attenuate low frequencies.

Golden Rule: The Impulse Invariance method **CANNOT** be used to design **Highpass, Bandstop, or Differentiator** filters. It is strictly suitable only for ****band-limited Lowpass and Bandpass filters**** where the stopband attenuation at Nyquist ($\Omega = \pi/T_s$) is already sufficiently deep ($> 30\text{--}40$ dB).

5 5. SYSTEMATIC STEP-BY-STEP DESIGN PROTOCOL

To design a digital IIR lowpass filter from continuous specifications:

1. **Map Digital Frequencies to Analog:** Using $\Omega = \omega/T_s$, compute:

$$\Omega_p = \frac{\omega_p}{T_s}, \quad \Omega_s = \frac{\omega_s}{T_s} \quad (14)$$

2. **Determine Analog Filter Order N and Cutoff Ω_c :** For a Butterworth prototype with passband ripple A_p (dB) and stopband attenuation A_s (dB):

$$N \geq \frac{\log_{10} \left(\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1} \right)}{2 \log_{10}(\Omega_s/\Omega_p)}, \quad \Omega_c = \frac{\Omega_p}{(10^{0.1A_p} - 1)^{1/(2N)}} \quad (15)$$

3. **Formulate Analog Transfer Function $H_a(s)$:** Place N poles symmetrically on the left-half circle of radius Ω_c :

$$s_k = \Omega_c e^{j \frac{\pi}{2N} (2k+N-1)}, \quad k = 1, 2, \dots, N \quad (16)$$

4. **Perform Partial Fraction Expansion:** Express $H_a(s)$ as a sum of simple 1st-order terms:

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k}, \quad \text{where } A_k = \lim_{s \rightarrow s_k} (s - s_k) H_a(s) \quad (17)$$

5. **Transform to Digital $H(z)$:** Replace each analog partial fraction using the mapping rule:

$$H(z) = \sum_{k=1}^N \frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}} \quad (18)$$

6. **Combine Conjugate Pairs into Real Biquads:** Combine complex conjugate terms to obtain strictly real denominator polynomials.
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6. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

6.1 Example 24.1: First-Order Analog to Digital IIR Transformation

Problem Statement: An analog lowpass filter is described by the transfer function:

$$H_a(s) = \frac{2}{s + 2} \quad (19)$$

Determine the digital filter transfer function $H(z)$ using the Impulse Invariance method with a sampling period of $T_s = 0.5$ seconds.

Solution:

Step 1: Identify Pole and Residue $H_a(s)$ has a single real pole at $s_1 = -2$ with residue $A_1 = 2$.

Step 2: Apply the Impulse Invariance Mapping Rule Using (8) with $T_s = 0.5$ s:

$$H(z) = \frac{T_s A_1}{1 - e^{s_1 T_s} z^{-1}} \quad (20)$$

Substituting $A_1 = 2$, $s_1 = -2$, and $T_s = 0.5$:

$$s_1 T_s = (-2)(0.5) = -1.0 \quad (21)$$

$$T_s A_1 = (0.5)(2) = 1.0 \quad (22)$$

$$e^{s_1 T_s} = e^{-1} \approx 0.367879 \quad (23)$$

Step 3: Formulate Digital Transfer Function

$$\mathbf{H}(z) = \frac{1}{1 - 0.3679z^{-1}} \quad (24)$$

Step 4: DC Gain Verification

- Analog DC Gain: $H_a(0) = \frac{2}{0+2} = 1.0$.
- Digital DC Gain: At $z = 1$:

$$H(1) = \frac{1}{1 - 0.3679} = \frac{1}{0.6321} \approx \mathbf{1.5819} \quad (25)$$

Notice that $H(1) \neq 1.0$ due to the discrete approximation of continuous integration by rectangular summation! If exact DC matching is desired, the gain is scaled by $(1 - e^{-1}) \approx 0.6321$.

6.2 Example 24.2: Second-Order Butterworth Filter Design with Complex Poles

Problem Statement: An analog second-order Butterworth lowpass filter with cutoff $\Omega_c = 1$ rad/s has the transfer function:

$$H_a(s) = \frac{1}{s^2 + \sqrt{2}s + 1} \quad (26)$$

Design the corresponding digital IIR filter using impulse invariance with $T_s = 1$ second.

Solution:

Step 1: Factor the Denominator to Find Analog Poles

$$s^2 + \sqrt{2}s + 1 = 0 \implies s_{1,2} = \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = -\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}} \approx -0.7071 \pm j0.7071 \quad (27)$$

Step 2: Partial Fraction Expansion

$$H_a(s) = \frac{1}{(s - s_1)(s - s_2)} = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} \quad (28)$$

where:

$$A_1 = \left. \frac{1}{s - s_2} \right|_{s=s_1} = \frac{1}{s_1 - s_2} = \frac{1}{j2(0.7071)} = \frac{1}{j\sqrt{2}} = -j\frac{1}{\sqrt{2}} \approx -j0.7071 \quad (29)$$

By complex conjugate symmetry, $A_2 = A_1^* = j\frac{1}{\sqrt{2}} \approx +j0.7071$.

Step 3: Map Poles to the Z-Domain With $T_s = 1$ s:

$$z_1 = e^{s_1 T_s} = e^{-0.7071 + j0.7071} = e^{-0.7071} [\cos(0.7071 \text{ rad}) + j \sin(0.7071 \text{ rad})] \quad (30)$$

Noting that $0.7071 \text{ rad} = 0.7071 \times \frac{180^\circ}{\pi} \approx 40.51^\circ$:

$$e^{-0.7071} \approx 0.49307 \quad (31)$$

$$\cos(40.51^\circ) \approx 0.76024, \quad \sin(40.51^\circ) \approx 0.64964 \quad (32)$$

$$z_1 = 0.49307(0.76024 + j0.64964) \approx \mathbf{0.3748 + j0.3203} \quad (33)$$

and $z_2 = z_1^* = \mathbf{0.3748 - j0.3203}$. Magnitude: $|z_1| = e^{-0.7071} = 0.4931 < 1$ ($\implies$ strictly stable!).

Step 4: Formulate Digital Partial Fractions

$$H(z) = \frac{T_s A_1}{1 - z_1 z^{-1}} + \frac{T_s A_2}{1 - z_2 z^{-1}} \quad (34)$$

Substituting $T_s = 1$, $A_1 = -j/\sqrt{2}$, $A_2 = +j/\sqrt{2}$:

$$H(z) = \frac{-j\frac{1}{\sqrt{2}}}{1 - z_1 z^{-1}} + \frac{j\frac{1}{\sqrt{2}}}{1 - z_1^* z^{-1}} \quad (35)$$

Putting over a common denominator:

$$\begin{aligned} H(z) &= \frac{-j\frac{1}{\sqrt{2}}(1 - z_1^* z^{-1}) + j\frac{1}{\sqrt{2}}(1 - z_1 z^{-1})}{(1 - z_1 z^{-1})(1 - z_1^* z^{-1})} \\ &= \frac{j\frac{1}{\sqrt{2}}(z_1^* - z_1)z^{-1}}{1 - 2\text{Re}(z_1)z^{-1} + |z_1|^2 z^{-2}} \end{aligned} \quad (36)$$

Evaluating the numerator:

$$z_1^* - z_1 = -j2\text{Im}(z_1) = -j2(0.3203) = -j0.6406 \quad (37)$$

$$j \frac{1}{\sqrt{2}}(z_1^* - z_1) = j \frac{1}{\sqrt{2}}(-j0.6406) = \frac{0.6406}{\sqrt{2}} \approx \mathbf{0.4530} \quad (38)$$

Evaluating the denominator:

$$2\text{Re}(z_1) = 2(0.3748) = \mathbf{0.7496} \quad (39)$$

$$|z_1|^2 = (0.49307)^2 \approx \mathbf{0.2431} \quad (40)$$

Step 5: Final Real Digital Biquad Transfer Function

$$H(z) = \frac{0.4530z^{-1}}{1 - 0.7496z^{-1} + 0.2431z^{-2}} \quad (41)$$

Notice that all coefficients are strictly real, and the poles lie comfortably inside the unit circle at radius $r = 0.4931$.

7 7. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Applying Impulse Invariance to the Whole Function Instead of Partial Fractions:** Students frequently try to substitute $s = \frac{1}{T_s} \ln(z)$ directly into $H_a(s)$. This produces a transcendental function, not a rational polynomial! Impulse invariance **must always be performed on partial fraction expansions:** $\frac{A}{s-s_k} \rightarrow \frac{TA}{1-e^{s_k T} z^{-1}}$.
- **Pitfall 2: Forgetting the Scaling Factor T_s :** Omitting T_s in the numerator produces a filter whose gain is magnified by $1/T_s$. For $T_s = 0.001$ s, the gain will be 1000× too high!
- **Pitfall 3: Believing Zero Locations Map via $z = e^{sT_s}$: Only poles map via $z_k = e^{s_k T_s}$. ZEROS DO NOT MAP via $z = e^{sT_s}$!** In Example 24.2, $H_a(s)$ had no finite zeros, yet $H(z)$ developed a zero at $z = 0$.
- **Pitfall 4: Attempting to Design a Highpass Filter via Impulse Invariance:** In semester exams, questions sometimes ask students to design a Highpass filter via Impulse Invariance. The astute student must state immediately that Impulse Invariance is fundamentally invalid for Highpass design due to catastrophic aliasing at DC.
- **Pitfall 5: Angle Unit Confusion in Handheld Calculators:** When computing $e^{s_k T_s} = e^{\sigma T_s} e^{j\Omega T_s}$, ΩT_s is in ****radians****, not degrees! Plugging radians into a calculator set to degree mode causes massive coefficient errors.

8 8. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 1: Impulse Invariance IIR Filter Design Quick Reference Sheet

Concept / Parameter	Mathematical Definition / Formula	Key Pedagogical Note
Invariance Criterion	$h[n] = T_s h_a(nT_s)$	Exact sampled analog impulse response
Pole Mapping Rule	$s_k \longrightarrow z_k = e^{s_k T_s}$	Single pole mapping
Partial Fraction Rule	$\frac{A_k}{s-s_k} \longrightarrow \frac{T_s A_k}{1-e^{s_k T_s} z^{-1}}$	Operational transformation recipe
Stability Condition	$\text{Re}(s_k) < 0 \implies z_k = e^{\text{Re}(s_k)T_s} < 1$	Stable analog filter $\implies$ Stable digital filter
Frequency Relation	$\omega = \Omega T_s$	Linear frequency mapping (before aliasing)
Digital Spectrum	$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega-2\pi k}{T_s}\right)$	Aliasing due to infinite frequency tails
Permissible Filters	Band-limited Lowpass and Bandpass	Stopband must attenuate sufficiently at π/T_s
Forbidden Filters	Highpass, Bandstop, Differentiator	Suffer severe aliasing distortion at DC
Mapping of Zeros	Zeros DO NOT map via $z = e^{sT_s}$	Only poles obey the exponential mapping